

Swift-Eye: Towards Anti-blink Pupil Tracking for Precise and Robust High-Frequency Near-Eye Movement Analysis with Event Cameras

Tongyu Zhang, Yiran Shen*, Guangrong Zhao, Lin Wang, Xiaoming Chen, Lu Bai* and Yuanfeng Zhou

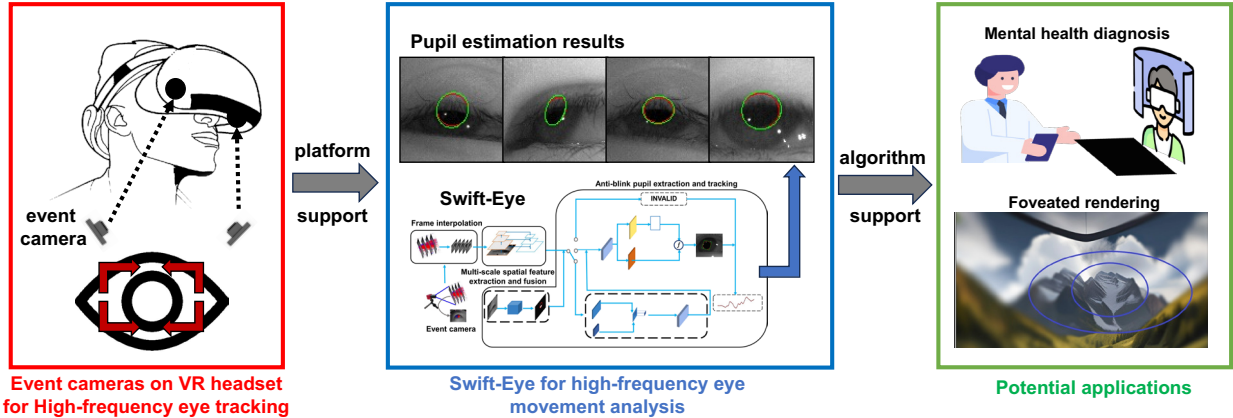


Fig. 1: Swift-Eye focuses on precise and robust pupil tracking to support high-frequency eye movement analysis with event cameras. It takes the low frame-rate video and high-temporal-resolution event stream as input and generates high-speed video recording of the eye movement. Then, the spatial feature and temporal features are fused to address the issue of occlusion over the pupil region to provide precise and robust pupil estimation. By fully uncovering the high temporal resolution of event cameras, Swift-eye would be the key enabler for the game-changing applications that require capturing the movement of human eyes in extremely high frequency, such as mental health diagnosis, foveated rendering, and so on.

Abstract—Eye tracking has shown great promise in many scientific fields and daily applications, ranging from the early detection of mental health disorders to foveated rendering in virtual reality (VR). These applications all call for a robust system for high-frequency near-eye movement sensing and analysis in high precision, which cannot be guaranteed by the existing eye tracking solutions with CCD/CMOS cameras. To bridge the gap, in this paper, we propose **Swift-Eye**, an offline precise and robust pupil estimation and tracking framework to support high-frequency near-eye movement analysis, especially when the pupil region is partially occluded. Swift-Eye is built upon the emerging event cameras to capture the high-speed movement of eyes in high temporal resolution. Then, a series of bespoke components are designed to generate high-quality near-eye movement video at a high frame rate over kilohertz and deal with the occlusion over the pupil caused by involuntary eye blinks. According to our extensive evaluations on EV-Eye, a large-scale public dataset for eye tracking using event cameras, Swift-Eye shows high robustness against significant occlusion. It can improve the IoU and F1-score of the pupil estimation by 20% and 12.5% respectively, compared with the second-best competing approach, when over 80% of the pupil region is occluded by the eyelid. Lastly, it provides continuous and smooth traces of pupils in extremely high temporal resolution and can support high-frequency eye movement analysis and a number of potential applications, such as mental health diagnosis, behaviour-brain association, etc. The implementation details and source codes can be found at <https://github.com/ztysdu/Swift-Eye>.

Index Terms—Eye tracking, event camera, feature fusion

1 INTRODUCTION

With the fast development of hardware and software design of Virtual Reality (VR), and the demand from various cutting-edge applications [12, 20], eye tracking has been enabled pervasively in current and

coming commercialized VR headsets as a new modality for human-computer interaction. However, current hardware and software solutions for eye tracking in VR only provide quite limited temporal resolution, typically around 30-90Hz, which constrains its applications for tasks requiring analyzing high-frequency eye movement in detailed phases. For example, mental health diagnosis [10, 13] requires very high-frequency eye trackers to analyze the high-speed movement of patients' eyes in high temporal resolution. Human eyes can move rapidly to achieve up to $700^\circ/s$ [4] angular speed and $24,000^\circ/s^2$ acceleration [5] in saccade status and last very short period. To support the detailed analysis of the dynamics of human eyes, the high-frequency eye tracking solution is required to uncover the information hidden in the erratic movement of human eyes and enable potential game-changing applications of VR headsets, such as mental health diagnosis.

However, existing mainstream eye tracking solutions rely on CCD/CMOS cameras, whose tracking frequency is limited by the frame rate of these cameras. To bridge the gap, the emerging bio-inspired vision sensors, namely event cameras [17], have been introduced to

- Tongyu Zhang, Yiran Shen, Guangrong Zhao and Yuanfeng Zhou are with School of Software, Shandong University.
- Lu Bai is with C-FAIR, Shandong University & Shandong Research Institute of Industrial Technology.
- Xiaoming Chen is with Beijing Technology and Business University.
- Lin Wang is with Hong Kong University of Science and Technology.
- Yiran Shen and Lu Bai are corresponding authors. Email: {yiran.shen, lubai}@sdu.edu.cn

Manuscript received 4 October 2023; revised 17 January 2024; accepted 24 January 2024. Date of publication 5 March 2024; date of current version 15 April 2024. Digital Object Identifier no. 10.1109/TVCG.2024.3372039

track the movement of human eyes in extremely high-frequency (*e.g.*, over tens of kilohertz) [8, 39]. Different from conventional CMOS/CCD cameras, an event camera produces an event stream consisting of asynchronous “events” encoding the locations and polarities of per-pixel intensity changes rather than synchronous video frames. An event is shot as soon as a per-pixel intensity change is detected; thus, the temporal resolution (*i.e.*, the most minor time difference between any consecutive events) of an event stream is high (tens of microseconds), making it ideal for capturing high-speed motion like the erratic eye movement. Besides the high temporal resolution, using event cameras for eye tracking also saves bandwidth and energy due to the sparse nature of events induced by near-eye movement [8]. Inspired by the advantages of event cameras above, Inivation Co.¹, the pioneer in designing and researching event cameras, has announced the extreme eye tracker [2], a high-frequency eye tracking peripheral with event cameras targeting at VR/AR applications and it will launch in the market in the near future. In contrast, though the high speed cameras are also capable of providing high frame-rate to support high-frequency tracking, it costs significant amount resources of computation, memory and bandwidth, which is too prohibitive for VR platforms.

Current studies on eye movement analysis with event cameras are emerging. For example, EVBEYE [8] proposes the first eye tracking algorithm with event cameras, which extracts and tracks the pupil region with a simple model-fitting and updating strategy. Then, EV-Eye [39] has collected a large-scale and diverse dataset to promote the research on pupil tracking in high temporal resolution. It utilizes the DAVIS346 event cameras [1] for data collection, which provides both the conventional video frames at a low frame rate (*i.e.*, 25fps) and event streams in high-temporal resolution. When building the data collection platform, it adopts the headrest to constrain the head movement, and two DAVIS346 cameras are placed near users’ eyes to simulate the eye tracking scenarios in VR/AR headsets. Though EV-Eye also proposes a new learning-based framework for pupil region estimation and tracking, the benchmarking algorithm evaluated along with the dataset focuses more on uncovering the potential of high-frequency tracking over kilohertz rather than accuracy. The robustness of the algorithm in estimating the pupil regions using event cameras is questionable especially when interference from other parts of the eyes is considered. For example, eye movement analysis can be interfered significantly by the occlusion of eyelids over the pupil region when the subjects are blinking involuntarily, leading to inaccurate and intermittent traces of pupil movement. Moreover, EV-Eye tracks the movement of pupils on the noisy and non-structural event streams directly, unlike the traditional video frames with prolific semantic information, they are poor in visualization for human perception as they lack sufficient texture details.

To bridge the gap above, in this paper, we propose **Swift-Eye**, an offline precise and robust near-eye pupil estimation and tracking framework to support extremely high-frequency eye movement analysis with event cameras. Swift-Eye features a number of bespoke components to deal with the issue of occlusion on the accuracy of pupil estimation. It employs the fine-tuned Timelens [35], a state-of-the-art event-based video interpolation method, to convert the event stream into frame-based video with a high frame-rate, *e.g.*, 5,000fps to support high-frequency near-eye movement visualization and further analysis. Then, a pupil estimation and tracking algorithm is proposed to estimate and track the pupil region through high-speed video. A temporal feature fusion component is designed to deal with the issue of occlusion. Also, a data synthesis approach with mask in-painting method, LAMA [33], is trained to generate more partially occluded eye images for data augmentation. In summary, the contributions of this paper are three-fold:

- We design and implement, Swift-Eye, a precise and robust pupil estimation and tracking framework, for near-eye movement visualization and analysis in extremely high frequency, by facilitating the high temporal resolution of event cameras.
- To deal with the issue of occlusion over the pupil region caused by

blinking eyelids, a temporal feature fusion component is designed to predict the occluded region and provide continuous tracking traces with high temporal resolution.

- We conduct extensive evaluations on the state-of-the-art and publicly-available event-based eye tracking dataset, EV-Eye [39], and compare with a number of state-of-the-art approaches. The results show that Swift-Eye achieves superior accuracy and robustness for pupil estimation and tracking tasks compared to the competing approaches. Specifically, it improves the IoU and F1-score of the pupil estimation by 20% and 12.5% respectively, compared with the second-best competing method when the occlusion ratio is over 80%.

2 RELATED WORK

This section introduces the related works, including eye tracking techniques, their broad applications and specific applications in VR.

2.1 Overview of eye tracking techniques

Eye tracking methods can be usually divided into two categories: model-based and appearance-based. Appearance-based gaze estimation methods directly regress from eye images to gaze directions or Points of Gaze (PoG), while the model-based method [18] predicts gazes using a geometric eye model. The model-based method is more relevant to our work and typically consists of two stages: identification of key eye characteristics and gaze regression. The process of identifying key eye characteristics involves analyzing the near-eye image and extracting salient characteristics of the eyes, such as pupils, irises, etc. This step is a prerequisite for gaze regression, and many studies focus on it.

Conventional eye tracking. The position of the pupil is one of the most important features of the eye. In the field of conventional eye tracking, drawing upon deep learning, segmentation-based pupil detection methods such as DeepVOG [38] and EyeNet [37] have been shown to outperform traditional pupil detection methods [16, 25]. Yu Feng et al. [14] proposed to use software-emulated events to determine the Region of Interest (ROI) on which the segmentation would be performed, achieving high speed. However, this approach did not utilize events to capture the high-frequency eye movements. These segmentation-based methods first classify each pixel in the image and then employ an ellipse-fitting algorithm [19] to get the outline of the pupil. However, such segmentation-based methods do not utilize the prior knowledge that the pupil is inherently elliptical. Therefore, they perform poorly when the pupil is partially occluded. More recently, EllSeg [23] is designed to regress the ellipse parameters of the pupil in an end-to-end fashion, leveraging the prior information that the pupil is an ellipse. This makes it more robust in scenarios involving partial occlusion of the pupil compared to segmentation-based methods. However, the end-to-end approach requires the model to regress the ellipse based on features from the entire image, which is coarse-grained, different from our fine-grained method that determines the ellipse based on a specific part of the feature map that the pupil is located in.

Event-based eye tracking. Limited by the hardware design of CCD/CMOS cameras, conventional eye tracking methods can only reach a few hundred hertz. Event-based eye tracking has been proposed to provide high-frequency eye tracking. EBVEYE [8] is the first event-based eye tracking algorithm; it uses grayscale images and subsequent events to obtain the ellipse of the pupil. Due to the lack of diversity in its datasets, EV-EYE [39] was proposed, featuring a diverse range of subjects in its datasets and a hybrid frame-event eye tracking method. However, its matching-based method constrains the shape of the pupil until the next frame arrives, which does not align with real-world conditions. Moreover, both events-based eye tracking methods utilize the noisy and non-structural event streams directly, without using the information from adjacent frames; thus, they are poor in visualization quality for further study.

2.2 Applications of Eye Tracking

Previous studies have shown the efficacy of eye tracking technology in analyzing gaze patterns, which are potential indicators for mental health

¹<https://inivation.com/>

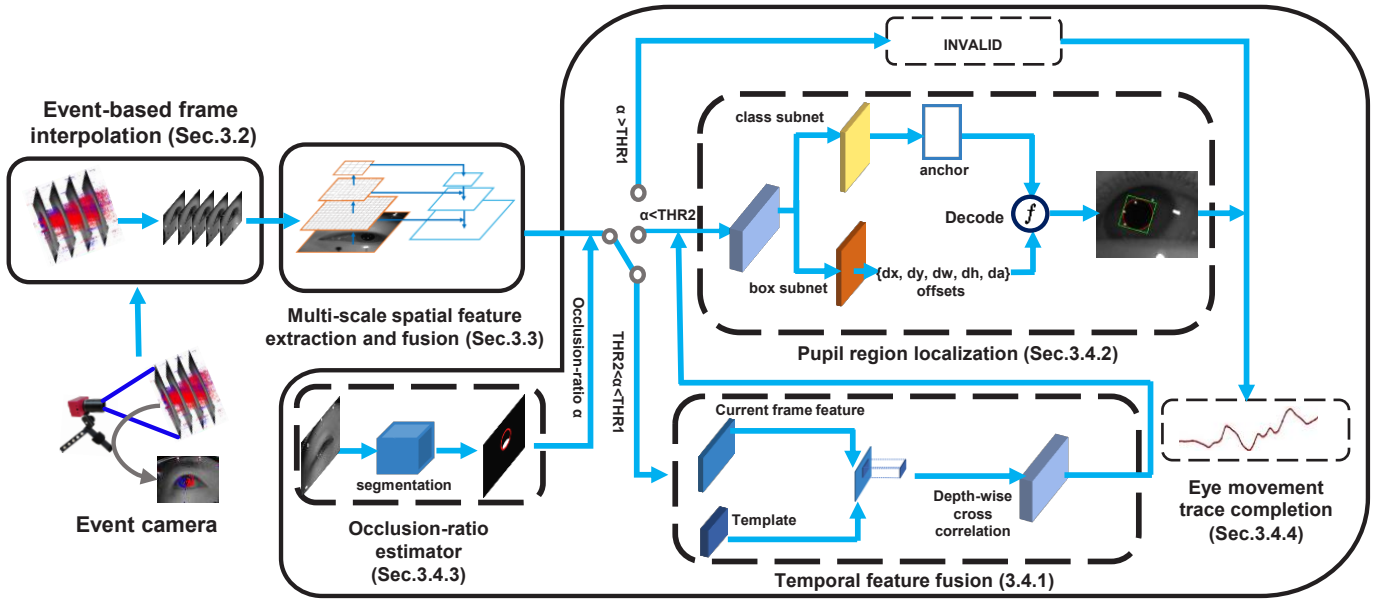


Fig. 2: **The framework of Swift-eye:** it consists of three major modules: event-based frame interpolation, multi-scale spatial feature extraction and fusion, and anti-blink pupil estimation and tracking.

conditions such as autism spectrum disorders [10, 13], amyotrophic lateral sclerosis (ALS), Alzheimer’s disease (AD), and Parkinson’s disease (PD) [34]. In addition, high-frequency eye movements have been found to provide richer information. Eckstein et al. [13] pointed out that eye tracking can complement both behavioral and brain measures. If eye tracking can be conducted at a high temporal resolution, it can capture how humans respond to tasks moment-by-moment. High-frequency eye movements also exhibit unique individual patterns, making the eye tracking method a feasible solution in authentication systems. This has been explored in studies [28, 30], both of which used the EyeLink 1000 eye tracker to capture eye movements and 1D-CNN-based neural networks [22] to extract patterns for authentication.

2.3 Eye Tracking in VR

Eye tracking has a wide range of applications in VR. First, it is efficient and convenient, making it promising for interaction in VR [29]. One study [15] proposed that eye input modality outperformed head input modality and was comparable to controller input in terms of throughput, adoption, and fatigue, highlighting its potential for next-generation AR/VR interactions. Another study [9] presented a hand-eye coordination method for efficiently and accurately selecting objects in 3D space subject to heavy occlusion.

In addition, eye tracking technology can ease the computing burden on VR devices. Foveated rendering [7] is a computational optimization method for VR that selectively renders only the portions of a scene where a user is currently gazing, requiring a high-frequency and accurate eye tracking method. A study [32] designed eye tracking and foveated rendering using approximation techniques, effectively reducing the overall cost of gaze-tracked foveated rendering while preserving user experience.

Moreover, eye tracking can evaluate learners’ cognitive load in a VR setting and assess their learning outcomes [6]. As proposed by Kiili et al. [21], a longer fixation duration reveals that a learner is under a high cognitive load. In comparison, a shorter fixation duration suggests that the learner is struggling to understand the content and is skimming over it. For assessing learning outcomes, one study [31] showed that average fixation duration could serve as an indicator to forecast learning outcomes.

3 DESIGN OF SWIFT-EYE

3.1 System overview

As shown in Fig. 2, our Swift-Eye consists of three bespoke modules for high-frequency eye movement visualization and analysis. Swift-Eye starts with an **event-based frame interpolation module** (Sec. 3.2), Timelens [35], to significantly increase the frame-rate of video recording (e.g., from 25fps to 5,000fps) by unveiling the potential of event cameras in high-frequency eye movement visualization and analysis. To guarantee the quality of interpolated frames for visualization, Timelens is fine-tuned in a self-supervised learning manner. Then, a **multi-scale feature extraction and fusion module** (Sec. 3.3) is implemented to extract and fuse semantic information from different scales by adopting the swin-transformer [27] and feature pyramid network (FPN) [26]. At last, an **anti-blink pupil estimation and tracking module** (Sec. 3.4) is proposed to deal with the interference caused by involuntary blinks during eye tracking. In this module, a temporal feature fusion component (Sec. 3.4.1) and a rotated pupil region detector (Sec. 3.4.2) are proposed to estimate and track the pupil region accurately through blinking eyes. An approach-adaptation mechanism with the occlusion-ratio estimator (Sec. 3.4.3) is designed to exclude unreliable prediction, and a simple linear interpolation strategy (Sec. 3.4.4) is adopted to obtain the complete trace for eye movement analysis and visualization.

Moreover, we propose a new data synthesis approach to save the efforts of annotating a large number of partially occluded eye images for training Swift-Eye and further improving its robustness against involuntary blinking. The approach utilizes and trains LAMA [33], a large mask in-painting tool, to synthesize partially occluded eye images from fully open ones. Then, the whole framework is trained with the fully-open eye images and partially-occluded synthesized eye images, and tested on real-world datasets with the issue of occlusion to demonstrate its accuracy and robustness.

3.2 Event-based frame interpolation

To effectively facilitate the high temporal resolution of event cameras for high-frequency eye movement analysis and visualization, our Swift-Eye adopts a state-of-the-art event-based frame interpolation network, Timelens [35], to convert the low frame-rate video (e.g., 25fps for DAVIS346) and asynchronous event streams into high-speed video (e.g., 5,000fps as in our experiment). Timelens takes a pair of consecutive frames and the in-between event stream as the inputs. By combining the high-quality dense optical flow estimated from the event stream and the

deep learning-based interpolation approach, it achieves state-of-the-art accuracy for a number of different challenging tasks and can adapt the frame rate of output conveniently as required.

Though Timelens is trained with several datasets in different environments, the outdoor scenes are far different from human eyes. Thus, we design a simple yet effective self-supervised finetuning approach to improve its interpolation quality for near-eye scenes. Specifically, a subset of EV-Eye [39] with low frame-rate video and the corresponding event streams are used to customize Timelens for near-eye movement analysis. During fine-tuning, each five consecutive frames and the in-between event streams are grouped together. The first and last frames are regarded as the input frames and the rest three frames are the ground-truth frames to be interpolated. Timelens takes the input frames and event stream to generate three synthetic frames. Corresponding losses can be estimated by comparing the difference between the synthetic frames and their ground truth. The whole fine-tuning process is self-supervised; no further manual efforts are needed to obtain the customized Timelens to interpolate near-eye images for movement analysis and visualization. By evaluating the EV-Eye dataset, we observe that the self-supervised finetuning on Timelens can improve the PSNR of the frame interpolation from 37.46dB to 41.16dB, showing 2.34 times improvement. The accurate interpolation can provide high-quality visualization of eye movement in detailed phases and support the following processing for high-frequency eye movement analysis.

3.3 Multi-scale spatial feature extraction and fusion

Extracting representative and robust features to encode the local-spatial information from the near-eye images is one of the key enablers for successful pupil region estimation. Thus, after the high-speed video of eye movement is obtained from the event-based frame interpolation module, each frame is fed into a multi-scale feature extraction and fusion module to generate hierarchical and integrated spatial features to support pupil estimation and tracking.

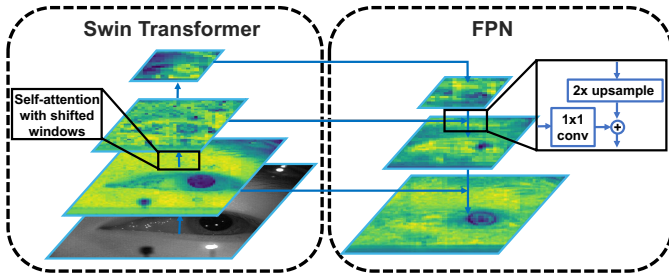


Fig. 3: Multi-scale spatial feature extraction and fusion via swin-transformer and FPN.

As shown in Fig. 3, the module starts with a swin-transformer [27] backbone for multi-scale feature extraction. The design of the swin-transformer is based on the prior knowledge that the object only occupies a fraction of the image rather than the whole. Therefore, it utilizes the shifted window for computing self-attention [36] within a window to extract the local spatial features while reducing the computational complexity. Then, from bottom to top, it merges adjacent patches to obtain higher-level features with a larger receptive field. As shown in the left part of Fig. 3, the hierarchical design of the architecture builds multi-scale spatial feature maps.

After the swin-transformer, a feature pyramid network (FPN) is applied to fuse the multi-scale spatial feature maps. As shown in the right part of Fig. 3, FPN is a top-down architecture with lateral connections. From top to bottom, the upper-layer feature map is up-sampled by two times and the lower-layer feature map goes through 1×1 convolution. Then, the processed feature maps from the two adjacent layers are added together to form a fused feature map that contains both rich details from the lower layer and high-level semantic information from the higher layers.

3.4 Anti-blink pupil estimation and tracking

After multi-scale features are extracted and fused, we propose a novel anti-blink pupil estimation and tracking module to deal with the occlusion caused by the user's involuntary blinking. The core idea is to design a temporal feature fusion component that takes an adaptive template obtained from the previous frame whose eye is fully open to assist the pupil estimation on the current frame when the pupil is partially occluded. We now describe the details.

3.4.1 Temporal feature fusion with adaptive template

When the pupil region is partially occluded due to involuntary blinking, it is challenging to estimate the correct region of the pupil based on the current image only. Fortunately, with the power of the event-based frame interpolation module (Sec. 3.2), we can obtain a high-speed video, *e.g.*, 5000fps, to observe the detailed phases of the near-eye movement, including the change of pupil, in a super-high temporal resolution, *e.g.*, 0.2ms. For each pair of consecutive frames, there should not be a noticeable difference between the pupil region, including size, location, and rotation. Thus, it is natural to exploit the temporal correlation through consecutive frames to address the problem of the partially occluded pupil.

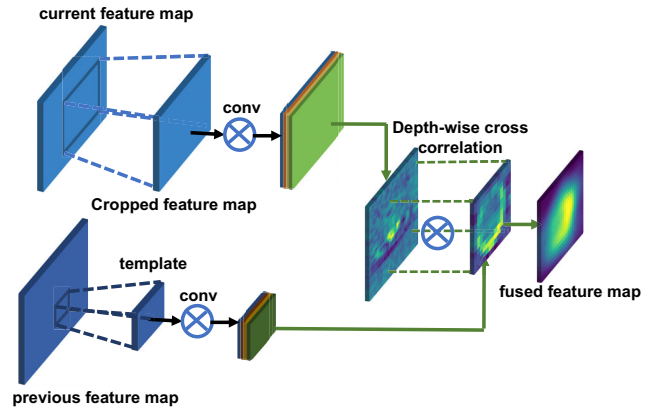


Fig. 4: Temporal feature fusion with an adaptive template obtained from the previous frame.

As shown in Fig. 4, our Swift-Eye maintains an adaptive template being updated through the video sequences. The template is obtained and updated from the previous frames whose eyes are fully open. For the current occluded eye image, its template is from the fully open eye image that is closest to it in the time domain. Specifically, as shown in Fig. 5, after the pupil region is localized successfully, a small square area centered at the middle of the pupil region is selected, *e.g.*, 52×52 pixels in our experiment, which contains the whole pupil region and its surrounding details. As the low-level local spatial information is crucial to distinguish the edge between the pupil and iris region, the corresponding area in the lowest level of the feature pyramid is used as the template (*i.e.*, a 13×13 square area *w.r.t* the selected 52×52 pixels in the image) for the temporal feature fusion in the next step.

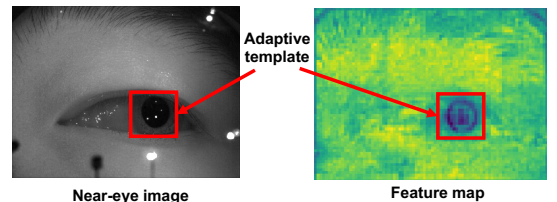


Fig. 5: Visualization of an example of an adaptive template in a near-eye image and the corresponding feature map.

Next, in the temporal feature fusion component, the adaptive template cropped from the feature pyramid of the previous frame is used to assist in the estimation of the partially occluded pupil region from the current frame. The template and the features extracted from the current frame are fused through the depth-wise cross-correlation operation [24]. It calculates the correlations between the template and the feature map of the current frame through depth-wise cross correlation [24] and integrates the results to form the fused feature map for pupil region localization in the next step. The detailed explanation can be referred to [24].

3.4.2 Pupil region estimation with rotated bounding box

With the aforementioned operations, a high-quality feature map is achieved by fusing the multi-scale features both in the spatial and temporal domains. A rotated pupil detector [40] is then applied to the feature map to estimate and track the pupil region through the video sequence.

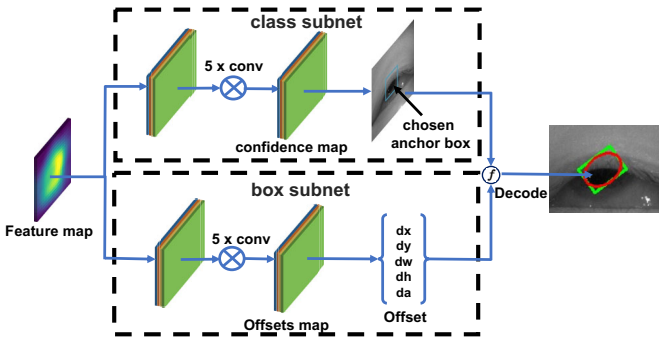


Fig. 6: The pupil region estimation with rotated bounding box to accommodate the rotation of the ellipse of the pupil.

It is known that the pupil region of the eyeball projected on the near-eye images is an ellipse. As shown in Fig. 6, the ellipse of the pupil region on the i_{th} image can be determined with five different parameters $\{P_x, P_y, P_w, P_h, P_a\}$, where P_x and P_y are the x-y pixel location of the ellipse center, P_w and P_h are the length of major and minor axes, and P_a is the rotation of the ellipse. It is obvious that the minimum bounding rectangle shares the same parameters with the ellipse; thus, it can be used to localize the pupil region.

This way, the pupil region localization task can be regarded as object (pupil) detection with a rotated bounding box. As shown in Fig. 6, the feature map is fed to a detection component, which consists of a class subnet and a box subnet. To realize accurate pupil localization with the rotated bounding box, numerous anchor boxes (e.g., 10,000 boxes in total in our experiment) with different sizes are generated beforehand. Then, the cropped features within each of the anchor box is fed to the detection component. The class subnet determines the confidence that the anchor box $\{B_x, B_y, B_w, B_h, B_a\}$ contains the pupil region, and the box subnet computes the offset $\{d_x, d_y, d_w, d_h, d_a\}$ from the corresponding anchor box to the bounding box of the pupil region. Lastly, the anchor box with the highest confidence and the corresponding offset is determined. As such, the final output through the rotated object detection can be calculated with the anchor box and corresponding offset using the decoding function below,

$$\begin{aligned} P_w &= \max\{B_w e^{d_w}, B_h e^{d_h}\} \\ P_h &= \min\{B_w e^{d_w}, B_h e^{d_h}\} \\ P_x &= d_x \cdot B_w \cos(B_a) - d_y \cdot B_h \sin(B_a) + B_x \\ P_y &= d_x \cdot B_w \sin(B_a) + d_y \cdot B_h \cos(B_a) + B_y \\ \alpha &= (B_a + d_a + \frac{\pi}{2}) \bmod \pi - \frac{\pi}{2} \\ \beta &= \begin{cases} \alpha & \text{if } B_w e^{d_w} > B_h e^{d_h} \\ \alpha + \frac{\pi}{2} & \text{otherwise} \end{cases} \\ P_a &= (\beta + \frac{\pi}{2}) \bmod \pi - \frac{\pi}{2} \end{aligned} \quad (1)$$

With the rotated bounding box, the pupil region can be determined by finding the largest ellipse inscribed in it.

3.4.3 Approach-adaptation via occlusion-ratio estimation

As per our observation and evaluation in Sec. 4.3, the pupil detector with temporal feature fusion shows superior performance on localizing partially occluded pupil regions. However, there are exceptional cases. Firstly, when the occlusion is too significant, e.g., over 90% occlusion, the correctness of the detected pupil region cannot be guaranteed, especially for fully closed eyes. Secondly, when there is no occlusion or it is minimal, e.g., less than 20%, the pupil detector without the temporal fusion component achieves higher accuracy as the template obtained from previous frames may also introduce interference on the pupil localization on the current frame.

According to the above discussion, we propose an adaptation scheme to switch among different approaches according to the ratio of the occlusion, i.e., the pupil detector without an adaptive template is chosen when the occlusion-ratio is less than 25% (THR1), the pupil detector with temporal feature fusion is adopted for occlusion-ratio between 25% to 87.5% (THR2), and at last, the results of the pupil localization are marked as **INVALID**, when the occlusion-ratio is over 87.5%. The parameters of pupils marked as **INVALID** will be interpolated in the following eye movement trace completion component. As estimating the ratio of occlusion of the current frame is difficult before localizing the whole pupil region and the difference between the occlusion-ratios of the adjacent frames should be minimal in the high-frequency video, we estimate the occlusion-ratio of the pupil based on the last frame rather than the current one to guide the approach-adaptation.

To estimate the occlusion-ratio, we adopt the U-Net-based pupil segmentation network introduced in [39] to segment out the disclosed area of pupil region to count the number of pixels of the area, termed as P_s . Then, the result of the pupil estimation in the last fully open frame indicates the whole area of the pupil, termed as P_l . Thus, the occlusion-ratio can be estimated as:

$$\alpha = \frac{P_l - P_s}{P_l} \quad (2)$$

This way, we can switch among different detection approaches according to the occlusion-ratio α and predefined thresholds.

3.4.4 Eye movement trace completion

As mentioned above, the pupil detector might give up estimating the parameters of the ellipse of the pupil region when the occlusion-ratio is too high (over 87.5% in our experiment), as the accuracy of the estimation degrades significantly and is not trustworthy. To obtain a complete trace of eye movement, we propose a simple yet effective approach to fill in the **INVALID** estimates via linear interpolation. For example, when a frame at timestamp t_i is marked as invalid. Our Swift-Eye searches for the most closely valid estimates before and after t_i , which, for example, are frames at timestamps t_j and t_k respectively. Thus, with the linear interpolation, we can obtain the estimates of the parameters $\Theta_i = \{x_i^c, y_i^c, a_i, b_i, \theta_i\}$ of an ellipse at timestamp t_i , i.e.,

$$\Theta_i = \frac{t_k - t_i}{t_k - t_j} \cdot \Theta_j + \frac{t_i - t_j}{t_k - t_j} \cdot \Theta_k \quad (3)$$

It is worth noting that, other interpolation mechanisms can be adopted as well, *e.g.*, spline interpolation, however, this is not the focus of our paper.

3.5 Data Synthesis with Mask In-painting

Training deep neural networks requires a significantly large number of near-eye images with labeled pupil regions. However, manually annotating a large number of partially occluded pupil regions is more challenging and time-consuming than fully-exposed ones. Moreover, picking up the near-eye images during blinking also costs significant effort as the blink normally lasts between 100ms to 400ms in about 2 to 10 seconds [3].

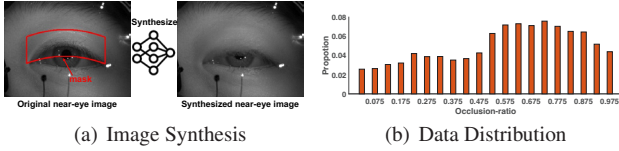


Fig. 7: Synthesizing partially occluded eye images using a trained LAMA [33] mask in-painting tool (a) and the distribution of occlusion ratios of the synthesized images.

To make life easier, we provide a simple yet effective data synthesis tool to enrich the training dataset with partially occluded near-eye images with labels. Different from labeling the pupil region manually, the data synthesis approach is based on LAMA [33], a mask-based image in-painting tool, to generate eye images with partially occluded pupil regions from the normal ones that have been already labeled in the original EV-Eye dataset [39]. The network of LAMA is re-trained using the near-eye images from EV-Eye to lead it to generate synthetic eyelids to cover part of the pupil region as shown in Fig. 7(a). Specifically, an eye detector is trained to localize the medial canthus (MC), lateral canthus (LC), and iris. Then, the edge curve of the eyelid can be approximated from these parts. During the training phase of LAMA, the eyelid is masked, and LAMA is trained to generate the eyelid. In the process of synthesizing partially occluded eye images, a quadratic curve formed by MC, LC, and a point in the pupil, along with its upper region, is masked, and LAMA inpaints this part as the eyelid, as shown in Fig. 7(a). The ratio of the occlusion is determined randomly to avoid over-fitting and its distribution is shown in Fig. 7(b). With the 9,011 fully open eye images labeled in the EV-Eye dataset, we generate the corresponding synthetic eye images with partially occluded pupil regions. At last, these pairs of near-eye images are used to train the parameters of our Swift-Eye. It is worth noting that, only the **synthetic** partially occluded face images are used in training. Although the edges of the generated eyelid and the eyelashes are still not as clear as in reality, as the evaluation results in Sec. 4, the model trained with synthetic images can be adopted directly to estimate and track the partially occluded pupil regions without any fine-tuning on **real** partially occluded eye images, which demonstrates its generalization in the real-world scenario.

4 EVALUATION

In this section, we will first describe the public dataset we use for evaluation and the extra labeling work on the partially occluded near-eye images. Then extensive quantitative and qualitative evaluations are conducted to justify the design of Swift-Eye through ablation studies and compare it with state-of-the-art pupil extraction/segmentation methods to show its accuracy and robustness on pupil estimation against blinking.

4.1 Dataset description

EV-Eye [39] provides a large-scale and diverse multi-modal frame-event dataset to promote event-based high-frequency eye tracking².

²<https://github.com/Ningreka/EV-Eye>

It consists of both low frame-rate video recording at 25fps and high-temporal-resolution event streams produced by two DAVIS346 event cameras placed close to the user's eyes to capture the eye movements. 48 users were recruited to participate in four data collection sessions. The first two sessions were conducted to collect the frame-event data produced by eye movements in saccade and fixation status and the rest two sessions were to collect the data caused by the eye movement in smooth pursuit. The dataset provides over 1.5 million near-eye grayscale images and more than 2.7 billion events for evaluating the methods on high-frequency eye movement tracking and analysis.

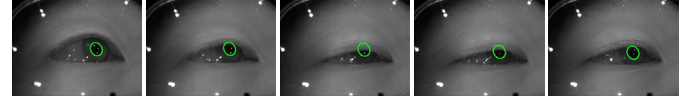


Fig. 8: A sequence of consecutive eye images with different ratios of occlusion with the manual annotations.

Extra Dataset labeling for evaluation. Though the original EV-Eye dataset has provided some labeled images for evaluation, the pupil regions of the labeled images are all fully-exposed, thus, are not sufficient for evaluating Swift-Eye which targets at alleviating the impact of occlusion caused by involuntary eye blinking. To bridge the gap, we manually annotate a number of near-eye images with different ratios of occlusion from EV-Eye dataset. For each user, 10 sequences are selected for annotation and each sequence consists of 5 consecutive frames capturing the eye movement while blinking, *i.e.*, from the frames with fully-exposed pupil regions until those are significantly occluded to cover pupil regions in different occlusion-ratios. Then the pupil regions of these selected eye images (2,400 in total) are manually annotated. To guarantee the correctness of the annotation, the shape consistency of pupil regions overtime is considered to infer the occluded part and the annotated ellipse should match the edge of the disclosed part. Finally, we obtain a subset of the annotated eye images with different ratios of occlusion for evaluating Swift-Eye. It is worth noting that, this annotated subset is not used to fine-tune the parameters of Swift-Eye, thus, can be used to prove the generalization of Swift-Eye trained with synthesized data. Fig. 8 illustrates a sequence of eye images while blinking from the EV-Eye dataset.

Dataset split. For training the pupil detector, the fully open eye images labeled in EV-Eye dataset along with the partially occluded eye images synthesized by LAMA are randomly split into ten folds. Nine folds are selected as training and the rest one is used as validation. The manually-annotated images in the extra dataset labeling are used as testing set for evaluation. Thus, there is no overlap between the eye images in training and testing set.

4.2 Implementation details and evaluation metrics

The training and evaluation of Swift-Eye are implemented on an NVIDIA GeForce RTX 3090 with 24GB VRAM using PyTorch 1.13. During training, the Adam optimizer is adopted, the learning rate is set as $1e-4$ and the batch size is set as 8. To achieve high-frequency eye tracking, the fine-tuned Timelens conducts event-based interpolation which converts the low frame-rate video (25fps) to high frame-rate video (5,000fps).

We use two popularly adopted metrics to evaluate the accuracy of pupil region estimation, which are,

IoU (Intersection over union) is commonly used to evaluate the accuracy of the pupil region segmentation [38], [23]. If the extracted pupil region is P and the ground truth is G , then the *IoU* can be expressed as,

$$IoU(P, G) = (P \cap G) / (P \cup G), \quad (4)$$

which estimates the overlap between the estimated region and ground truth region, *i.e.*, $IoU = 1$ indicates a perfect match while $IoU = 0$ means there is no overlap between the two regions.

F1-score (Dice coefficient) is the other popularly used evaluation metric for segmentation tasks. It can be expressed as,

$$F1(P, G) = (2|P \cap G|) / (|P| + |G|), \quad (5)$$

where $|P|$ and $|G|$ count the number of pixels in the two regions. The F1-score estimates the similarity between the estimated pupil region and ground truth, which ranges between 0 to 1, *i.e.*, a higher F1-score is desired.

4.3 Ablation studies

The evaluation starts from the ablation studies to determine the key parameter, *i.e.*, thresholds of the approach-adaptation component and justify the design of Swift-Eye by calculating the contributions of the key components to the accuracy of pupil region estimation.

4.3.1 Threshold of occlusion-ratio

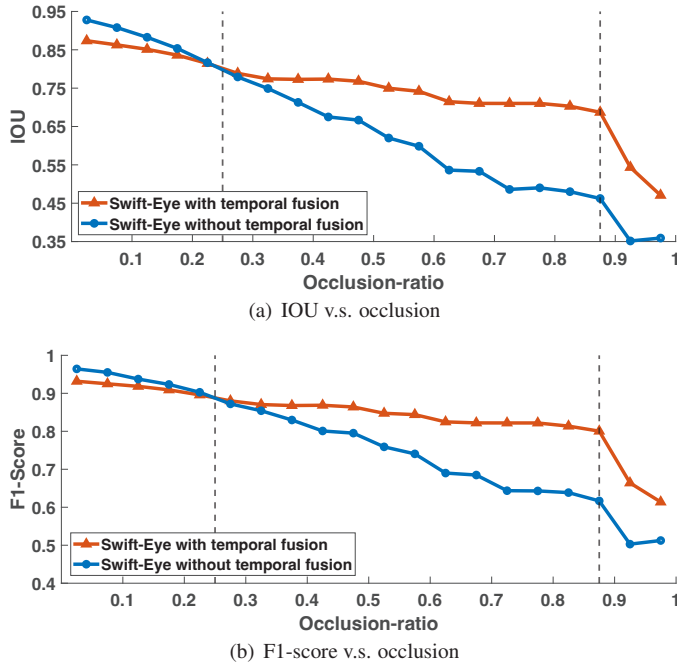


Fig. 9: The choice of thresholds.

As discussed in Sec. 3.4.3, different approaches should be chosen with different ratios of occlusion to achieve optimal performance of accuracy on pupil region estimation. The upper and lower thresholds of the occlusion-ratio are the key parameters to be determined as they control the approach-adaptation mechanism in the anti-blink pupil estimation and tracking module.

To determine the thresholds, we evaluate and compare the accuracy of Swift-Eye with or without the temporal feature fusion component under different ratios of occlusion. To guarantee a fair comparison, the two approaches are all trained with the synthetic partially occluded eye images and their original correspondence from the EV-Eye dataset while the evaluation is conducted on the real partially occluded eye image sequences annotated by ourselves. All the rest of the evaluations follow the same protocol for training and testing, unless otherwise stated. Fig. 9 presents the IOUs and F1-scores of the two approaches on estimating the pupil regions of the eye images under different occlusion-ratios. From the results, we can observe, both the IOUs and F1-scores of the two approaches decrease with the growth of the occlusion-ratio as the pupil estimation is growing challenging. By comparing between the two approaches, Swift-Eye with temporal feature fusion achieves significantly higher IOUs and F1-scores than that without the temporal feature fusion when the occlusion is considerable. Moreover, two key points can be observed from the figure to guide the choice of the thresholds. First, the accuracy of Swift-Eye with temporal feature fusion surpasses its counterpart when the occlusion-ratio goes over 0.25. The gap between the two approaches grows rapidly with the increase of occlusion-ratio. Then the occlusion-ratio keeps growing and reaches the second key point over 0.875, the accuracy of both of

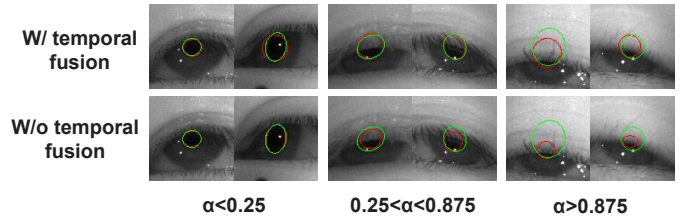


Fig. 10: Examples of pupil estimation results w/ or w/o temporal feature fusion around the chosen thresholds. The green ellipses are the manual annotations and red ellipses are the estimation results.

the approaches drops dramatically, which indicates the pupil detector cannot provide trustworthy estimation results. Therefore, we choose 0.25 as the lower threshold and 0.875 as the upper threshold, *i.e.*, when the occlusion-ratio is below 0.25, the temporal feature fusion is skipped as the template from the previous frame brings inaccuracy in the pupil region estimation. Then when the occlusion-ratio is between 0.25 and 0.875, the full-version Swift-Eye is adopted to facilitate temporal correlation information provided by the adaptive template to assist the pupil region estimation on the current frame. At last, when the occlusion-ratio is above 0.875, Swift-Eye skips this frame as it fails to provide trustworthy estimation results.

To further justify the choice of the upper and lower thresholds, we show some examples of near-eye images with occlusion-ratios around the chosen thresholds (before and after the thresholds) in Fig. 10. The ground truth (green ellipses) and estimation results (red ellipses) of Swift-Eye with or without temporal feature fusion are shown in different rows. From the figures we can confirm our choice of thresholds, *i.e.*, Swift-Eye without temporal fusion is chosen for occlusion-ratio less than 0.25 and Swift-Eye with temporal fusion is an apparent choice when the occlusion-ratio is over 0.25. At last, the accuracy of the pupil estimation experiences a significant drop when the occlusion-ratio is over 0.875, thus, it is not trustworthy anymore.

4.3.2 Contributions of the key components

Swift-Eye includes two key components to address the issue of occlusion caused by eye blinking, which are the temporal feature fusion with adaptive template introduced in Sec. 3.4.1 and the data synthesis with mask in-painting described in Sec. 3.5. To justify the design of Swift-Eye and evaluate the contributions of these two key components to the robustness of Swift-Eye, we successively remove one of the components and compute the accuracy of the remaining framework on pupil region estimation against different occlusion-ratios.

Fig. 11 presents the evaluation results of the full-version Swift-Eye, the Swift-Eye without temporal feature fusion component and the Swift-Eye without both temporal feature fusion and data synthesis, respectively. From the results, we can find the growth of the occlusion-ratio poses significant challenges to pupil estimation and leads to a decrease in both IOUs and F1-scores of all the three approaches. However, the impact of the occlusion on the accuracy is significantly different in different approaches. The full-version Swift-Eye demonstrates superior robustness against the growth of occlusion-ratio, both IOUs and (especially) F1-scores decrease quite gracefully. The comparison between Swift-Eye with or without temporal feature fusion demonstrates that the proposed temporal feature fusion component contributes significantly to the robustness of Swift-Eye against partial occlusion over the pupil region. At last, without both temporal feature fusion and the synthesized eye images from mask in-painting, Swift-Eye is very sensitive to the change of occlusion and its IOUs and F1-scores drop significantly with growing occlusion-ratio. From the discussion above, we can claim that both the two components make significant contributions to the robustness of Swift-Eye.

We further confirm the contributions of the key components by providing qualitative evaluation results of the pupil estimation under different occlusion-ratios. Fig. 12 presents the examples from the three different approaches. By comparing through different rows, we

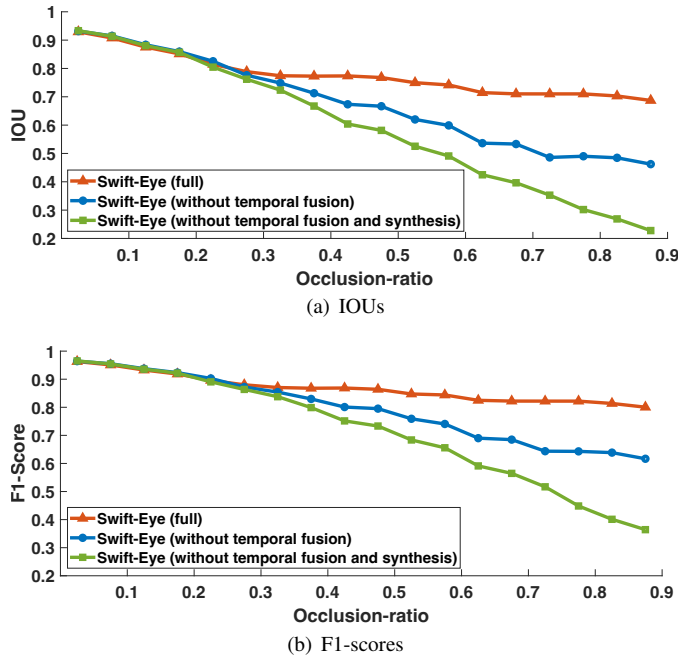
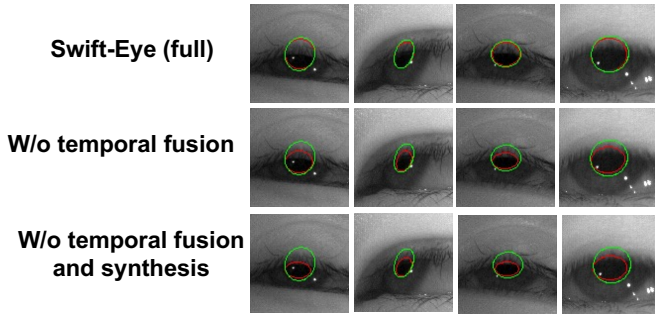


Fig. 11: The contributions of the proposed key components.

Fig. 12: Examples of pupil estimation results w/ or w/o some key components. The **ellipse** are ground truth annotations and **red** ellipses are the estimation results.

can observe clear improvement by progressively adding the two key components. After the ablation studies, we can confirm the design of Swift-Eye and determine the values of the thresholds to achieve the highest accuracy on pupil estimation and tracking against various ratios of occlusion.

4.4 Comparison with state-of-the-art approaches

We then evaluate and compare Swift-Eye with three different state-of-the-art pupil estimation or tracking approaches including EV-Eye [39], DeepVoG [38] and EllSeg [23]. EV-Eye is currently the state-of-the-art method for eye tracking with event cameras while DeepVoG and EllSeg work only on video frames.

Competing methods. **EV-Eye** is a new benchmarking method open-sourced with the EV-Eye public dataset. It consists of a U-Net-based segmentation network for pupil estimation on the near-eye images. Then the segmented pupil region is abstracted as a template and is tracked through the event stream to provide high-frequency eye tracking. The segmentation network only estimates the disclosed region of the pupil, thus, cannot deal with the problem of occlusion. **DeepVoG** shares a similar backbone to EV-Eye, i.e., It also uses a structure similar to U-Net for segmentation. While, after the segmentation, DeepVoG employs a regression function to take the segmentation results as input and generate an ellipse. The second difference is that EV-Eye contains an

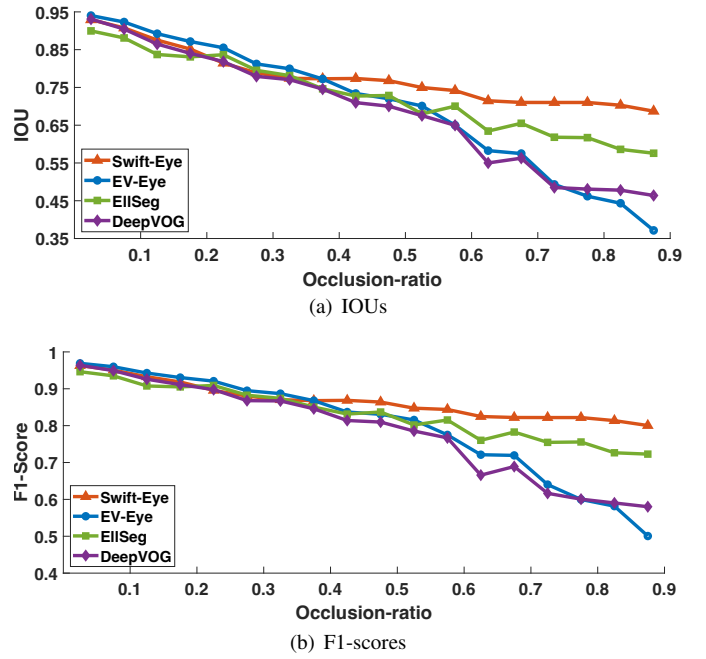
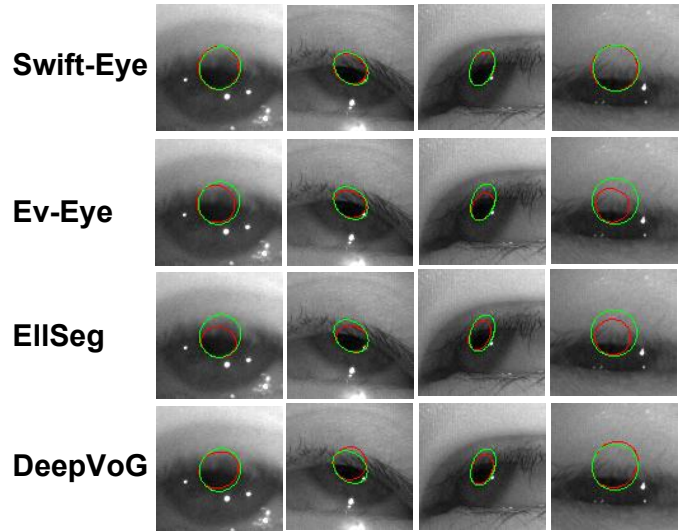


Fig. 13: The comparison among state-of-the-art pupil localization approaches.

event-based tracking component to achieve high-frequency eye tracking through event streams while DeepVoG only works on video frames. **EllSeg** also considers the issue of occlusion caused by involuntary blinks. It employs an encoder-decoder architecture as its backbone. The encoder extracts features of the near-eye images and maps them to the latent representation. Then the decoder regresses the parameters of an ellipse fitting on the pupil region.

Fig. 14: Examples of pupil estimation results of Swift-Eye, EV-Eye, EllSeg and DeepVoG. The **ellipse** are ground truth annotations and **red** ellipses are the estimation results.

Evaluation results. Swift-Eye and the three competing methods are all trained using the synthetic eye images and their original correspondences. The annotated subset containing near-eye images with various ratios of occlusion is used solely in evaluation. To guarantee fair comparison, DeepVoG and EllSeg are applied on the high frame-rate (5,000fps) video generated from the fine-tuned Timelens adopted in

Swift-Eye for event-based frame interpolation and the weights of all models are initialized randomly without any pretraining.

Fig. 13 presents the IoUs and F1-scores of the four methods on pupil estimation over different occlusion-ratios. It is clear that the occlusion over the pupil region has a negative impact on the accuracy of pupil estimation for all four methods. Then by comparing among different methods, our proposed Swift-Eye achieves the best performance on pupil estimation especially when the occlusion-ratio is high. Specifically, the IoUs and F1-scores of Swift-Eye achieve 20% (from 0.58 to 0.702) and 12.5% (from 0.726 to 0.817) improvement respectively, over the second best approach, EllSeg. Then EllSeg achieves the second place as it also considers the issue of occlusion. At last, DeepVOG and EV-Eye are quite close to each other on pupil estimation because they share almost the same backbone component, a U-Net network for pupil region segmentation.

To provide a more intuitive understanding of the quality of the pupil estimation of different methods, we present a number of examples of near-eye images with estimated pupil regions obtained from different methods under different occlusion-ratios arranged in different rows of Fig. 14. By comparing the pupil estimation results of different methods, intuitively we can find Swift-Eye provides the best results, which better match the ground truth annotation than the results from other methods.

4.5 Eye movement trace in high temporal resolution

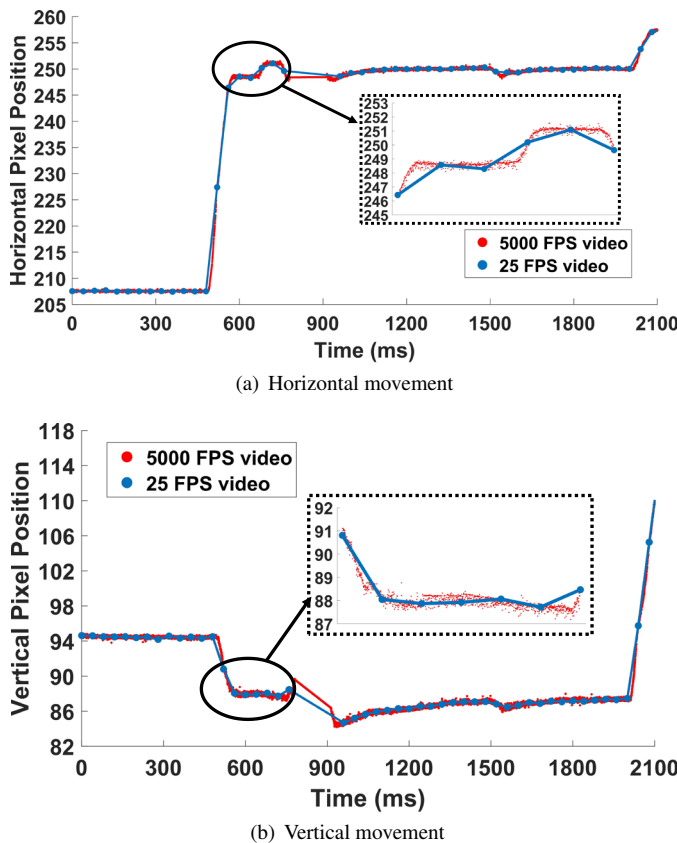


Fig. 15: High-frequency and low-frequency traces of eye movement (pupil center) in horizontal and vertical directions

At last, we demonstrate the benefits of tracking eye movement in high temporal resolution by comparing the traces obtained from the pupil estimation on videos with different frame rates: the low-speed video at 25fps and the high-speed video at 5,000fps. The low-speed video is the original video recording from the EV-Eye dataset and the high-speed video is generated from event-based frame interpolation.

Fig. 15 presents an example of the horizontal and vertical traces of the eye movement (i.e., the locations of pupil center) estimated by

Swift-Eye on a 2-second video clip. The blue dots are from the original video and the red dots are from the high-speed video generated by the fine-tuned Timelens. From the figure we can find, though the overall trend of the two traces tend to be similar, undeniable difference can be observed when we zoom in on the local details of the traces. For example, from the zoomed view of Fig. 15(a), we can clearly see the detailed eye movement in high-speed video from the dense red dots which are different from simply interpolating the discrete blue dots obtained from low frame-rate video. These discrepancies normally happen when the eye is fast moving, thus requiring high-frequency tracking techniques to reveal the imperceptible knowledge behind the eye movement. Therefore, our proposed Swift-Eye can well-support the high-frequency eye movement analysis and further provide important hints for high-level applications like mental health diagnosis [10, 11] or research on the association between behavior (eye movement) and brain mechanism [13].

5 CONCLUSION AND FUTURE WORK

In this paper, we proposed Swift-Eye, a precise and robust pupil region estimation and tracking framework to support the high-frequency near-eye movement analysis with event cameras targeting at VR applications. It started from an event-based frame interpolation module to generate high-speed videos (e.g., 5,000fps) from the synchronized low frame-rate videos (25fps) and the event streams to provide high temporal resolution and high-quality visualization for high-frequency eye movement analysis. The eye movement analysis relied on an anti-blink pupil estimation and tracking module to address the issue of occlusion caused by involuntary blinking. To improve the robustness of Swift-Eye against occlusion, a temporal feature fusion component was proposed to incorporate the features from an adaptive template extracted from previous frames to assist the pupil estimation of the current frame. Then an approach-adaptation component was designed to switch among different pupil detection strategies to achieve optimal accuracy. According to extensive quantitative and qualitative evaluations on the large-scale EV-Eye dataset, Swift-Eye outperformed the state-of-the-art methods significantly.

It is worth noting that, our current work did not consider the issue of tracking latency. Swift-Eye focused on providing accurate and robust eye movement traces in high temporal resolution, which was more appropriate for applications requiring only offline eye movement analysis, such as mental health diagnosis and behavior-brain association, rather than requiring low-latency tracking, such as foveated rendering. In the future, we will work on analyzing the tracking latency of Swift-Eye and improving the efficiency of the algorithms to uncover its potential as a real-time eye tracking method.

ACKNOWLEDGMENTS

This work was supported partially by National Natural Science Foundation of China under Grant No. 62177001 and 62371273; Natural Science Foundation of Shandong Province, Grant No. 2022HWYQ-040 and ZR2023YQ058; Taishan Scholars Program; Beijing Natural Science Foundation under Grant 4222003.

REFERENCES

- [1] Davis346 event camera. <https://inivation.com/wp-content/uploads/2019/08/DAVIS346.pdf>, 2
- [2] Extreme eye tracking with event cameras. <https://www.foveator.ai/>, 2
- [3] Wikipedia blinking. <https://en.wikipedia.org/wiki/Blinking>, 6
- [4] Wikipedia saccade. <https://en.wikipedia.org/wiki/Saccade>, 1
- [5] R. A. Abrams, D. E. Meyer, and S. Kornblum. Speed and accuracy of saccadic eye movements: characteristics of impulse variability in the oculomotor system. *Journal of Experimental Psychology: Human Perception and Performance*, 15(3):529, 1989. 1
- [6] I. B. Adhanom, P. MacNeilage, and E. Folmer. Eye tracking in virtual reality: A broad review of applications and challenges. *Virtual Reality*, pp. 1–25, 2023. 3
- [7] R. Albert, A. Patney, D. Luebke, and J. Kim. Latency requirements for foveated rendering in virtual reality. *ACM Transactions on Applied Perception (TAP)*, 14(4):1–13, 2017. 3

- [8] A. N. Angelopoulos, J. N. Martel, A. P. Kohli, J. Conradt, and G. Wetzstein. Event-based near-eye gaze tracking beyond 10,000 Hz. *IEEE Transactions on Visualization and Computer Graphics*, 27(5):2577–2586, 2021. 2
- [9] Y. Bao, J. Wang, Z. Wang, and F. Lu. Exploring 3d interaction with gaze guidance in augmented reality. In *2023 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*, pp. 22–32, 2023. doi: 10.1109/VR55154.2023.00018 3
- [10] Z. Boraston and S.-J. Blakemore. The application of eye-tracking technology in the study of autism. *The Journal of Physiology*, 581(3):893–898, 2007. doi: 10.1113/jphysiol.2007.133587 1, 3, 9
- [11] A. H. Clarke, J. Ditterich, K. Drüen, U. Schönfeld, and C. Steineke. Using high frame rate cmos sensors for three-dimensional eye tracking. *Behavior Research Methods, Instruments, & Computers*, 34(4):549–560, 2002. 9
- [12] A. T. Duchowski. A breadth-first survey of eye-tracking applications. *Behavior Research Methods, Instruments, & Computers*, 34(4):455–470, 2002. 1
- [13] M. K. Eckstein, B. Guerra-Carrillo, A. T. M. Singley, and S. A. Bunge. Beyond eye gaze: What else can eyetracking reveal about cognition and cognitive development? *Developmental Cognitive Neuroscience*, 25:69–91, 2017. 1, 3, 9
- [14] Y. Feng, N. Goulding-Hotta, A. Khan, H. Reyserhove, and Y. Zhu. Real-time gaze tracking with event-driven eye segmentation. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 399–408. IEEE, 2022. 2
- [15] A. S. Fernandes, T. S. Murdison, and M. J. Proulx. Leveling the playing field: A comparative reevaluation of unmodified eye tracking as an input and interaction modality for vr. *IEEE Transactions on Visualization and Computer Graphics*, 29(5):2269–2279, 2023. doi: 10.1109/TVCG.2023.3247058 3
- [16] W. Fuhl, T. Kübler, K. Sippel, W. Rosenstiel, and E. Kasneci. Excuse: Robust pupil detection in real-world scenarios. In *Computer Analysis of Images and Patterns: 16th International Conference, CAIP 2015, Valletta, Malta, September 2-4, 2015 Proceedings, Part I 16*, pp. 39–51. Springer, 2015. 2
- [17] G. Gallego, T. Delbrück, G. Orchard, C. Bartolozzi, B. Taba, A. Censi, S. Leutenegger, A. J. Davison, J. Conradt, K. Daniilidis, et al. Event-based vision: A survey. *IEEE transactions on pattern analysis and machine intelligence*, 44(1):154–180, 2020. 1
- [18] E. D. Guestrin and M. Eizenman. General theory of remote gaze estimation using the pupil center and corneal reflections. *IEEE Transactions on biomedical engineering*, 53(6):1124–1133, 2006. 2
- [19] R. Halir and J. Flusser. Numerically stable direct least squares fitting of ellipses. In *Proc. 6th International Conference in Central Europe on Computer Graphics and Visualization. WSCG*, vol. 98, pp. 125–132. Citeseer, 1998. 2
- [20] K. Harezlak and P. Kasprowski. Application of eye tracking in medicine: A survey, research issues and challenges. *Computerized Medical Imaging and Graphics*, 65:176–190, 2018. 1
- [21] K. Kiili, H. Ketamo, and M. D. Kickmeier-Rust. Eye tracking in game-based learning research and game design. *International Journal of Serious Games*, 1(2):51–65, 2014. 3
- [22] S. Kiranyaz, O. Avci, O. Abdeljaber, T. Ince, M. Gabbouj, and D. J. Inman. 1d convolutional neural networks and applications: A survey. *Mechanical systems and signal processing*, 151:107398, 2021. 3
- [23] R. S. Kothari, A. K. Chaudhary, R. J. Bailey, J. B. Pelz, and G. J. Diaz. Ellseg: An ellipse segmentation framework for robust gaze tracking. *IEEE Transactions on Visualization and Computer Graphics*, 27(5):2757–2767, 2021. 2, 6, 8
- [24] B. Li, W. Wu, Q. Wang, F. Zhang, J. Xing, and J. Yan. Siamrpn++: Evolution of siamese visual tracking with very deep networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 4282–4291, 2019. 5
- [25] D. Li, D. Winfield, and D. J. Parkhurst. Starburst: A hybrid algorithm for video-based eye tracking combining feature-based and model-based approaches. In *2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR'05)-Workshops*, pp. 79–79. IEEE, 2005. 2
- [26] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie. Feature pyramid networks for object detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 2117–2125, 2017. 3
- [27] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 10012–10022, 2021. 3, 4
- [28] D. Lohr, H. Griffith, and O. V. Komogortsev. Eye know you: Metric learning for end-to-end biometric authentication using eye movements from a longitudinal dataset. *IEEE Transactions on Biometrics, Behavior, and Identity Science*, 4(2):276–288, 2022. 3
- [29] P. Majaranta and A. Bulling. Eye tracking and eye-based human–computer interaction. In *Advances in physiological computing*, pp. 39–65. Springer, 2014. 3
- [30] S. Makowski, L. A. Jäger, P. Prasse, and T. Scheffer. Biometric identification and presentation-attack detection using micro-and macro-movements of the eyes. In *2020 IEEE International Joint Conference on Biometrics (IJCB)*, pp. 1–10. IEEE, 2020. 3
- [31] N. A. Rappa, S. Ledger, T. Teo, K. Wai Wong, B. Power, and B. Hilliard. The use of eye tracking technology to explore learning and performance within virtual reality and mixed reality settings: a scoping review. *Interactive Learning Environments*, 30(7):1338–1350, 2022. 3
- [32] R. Singh, M. Huzaifa, J. Liu, A. Patney, H. Sharif, Y. Zhao, and S. Adve. Power, performance, and image quality tradeoffs in foveated rendering. In *2023 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*, pp. 205–214. IEEE, 2023. 3
- [33] R. Suvorov, E. Logacheva, A. Mashikhin, A. Remizova, A. Ashukha, A. Silvestrov, N. Kong, H. Goka, K. Park, and V. Lempitsky. Resolution-robust large mask inpainting with fourier convolutions. *arXiv preprint arXiv:2109.07161*, 2021. 2, 3, 6
- [34] L. Tao, Q. Wang, D. Liu, J. Wang, Z. Zhu, and L. Feng. Eye tracking metrics to screen and assess cognitive impairment in patients with neurological disorders. *Neurological Sciences*, 41:1697–1704, 2020. 3
- [35] S. Tulyakov, D. Gehrig, S. Georgoulis, J. Erbach, M. Gehrig, Y. Li, and D. Scaramuzza. Time lens: Event-based video frame interpolation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 16155–16164, 2021. 2, 3
- [36] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017. 4
- [37] Z. Wu, S. Rajendran, T. van As, J. Zimmermann, V. Badrinarayanan, and A. Rabinovich. Eynet: A multi-task network for off-axis eye gaze estimation and user understanding. *arXiv preprint arXiv:1908.09060*, 2019. 2
- [38] Y.-H. Yiu, M. Aboulatta, T. Raiser, L. Ophey, V. L. Flanagan, P. Zu Eulenburg, and S.-A. Ahmadi. Deepvov: Open-source pupil segmentation and gaze estimation in neuroscience using deep learning. *Journal of neuroscience methods*, 324:108307, 2019. 2, 6, 8
- [39] G. Zhao, Y. Yang, J. Liu, N. Chen, Y. Shen, H. Wen, and G. Lan. Ev-eye: Rethinking high-frequency eye tracking through the lenses of event cameras. In *Thirty-seventh Conference on Neural Information Processing Systems(NeurIPS)*, New Orleans, USA, December 2023. 2, 4, 5, 6, 8
- [40] Y. Zhou, X. Yang, G. Zhang, J. Wang, Y. Liu, L. Hou, X. Jiang, X. Liu, J. Yan, C. Lyu, W. Zhang, and K. Chen. Mmrotate: A rotated object detection benchmark using pytorch. In *Proceedings of the 30th ACM International Conference on Multimedia*, 2022. 5